CLOUD VS ON-PREMISE

ON-PREMISES

Description

- Physical servers and storage systems owned and maintained by the organization.
- Complete control over hardware and software configurations.
- Data stays within the organization's physical boundaries.

Advantages

- Full control over security and compliance.
- Predictable performance (no cloud network latency).
- No data egress costs.
- Suitable for highly sensitive data with strict regulatory requirements.

Challenges

- High upfront capital expenditure (CapEx).
- Limited scalability — adding capacity requires buying new hardware.
- Maintenance overhead (hardware failures, upgrades, patches).
- Hard to handle variable workloads efficiently.
- Slow provisioning of new resources.

CLOUD

Description

- Resources provided as services over the internet.
- Pay-as-you-go (OpEx) pricing model.
- Managed by cloud service providers.

Advantages

- **Elasticity:** scale resources up or down based on demand.
- **Cost efficiency:** convert CapEx to OpEx; pay only for usage.
- **Speed:** provision resources in minutes.
- **Global reach:** deploy pipelines close to sources or users.

- **Managed services:** focus on engineering, not infrastructure.
- **Innovation:** fast access to AI/ML, streaming, analytics tools.

Challenges

- Data transfer (egress) costs.
- Potential vendor lock-in.
- Learning curve for cloud-specific tools.
- Security concerns if misconfigured.
- Compliance considerations depending on industry.

KEY CONSIDERATIONS FOR DATA ENGINEERING

1. Data Gravity

- Where your data lives matters.
- Moving large datasets between clouds is expensive and slow.
- Choose services close to your data sources.

2. Cost Optimization

- Understand pricing (compute, storage, transfer).
- Use spot/preemptible instances for batch jobs.
- Implement data lifecycle policies.

3. Performance Requirements

- Consider latency for real-time workloads.
- Compare managed vs. self-managed solutions.

4. Skills & Learning Curve

- Each cloud has its own architecture and tools.
- Consider training and hiring requirements.

5. Multi-Cloud Strategy

- Reduce lock-in risk.
- Use cloud-agnostic technologies (Kubernetes, Spark).
- Think about portability from day one.

CLOUD SERVICES COMPARISON

ENVIRONMENTAL IMPACT – ENVIRONMENTAL CONSIDERATIONS

Energy Consumption & Carbon Footprint

- Data centers consume significant electricity for computing and cooling.
- Geographic region affects carbon intensity (renewables vs. fossil fuels).
- Resource utilization efficiency directly affects environmental cost.

PRACTICE AWS

Lambda = comparable to:

- Cloud Run Functions (GCP)
- Azure Functions (Microsoft)

Details:

- Serverless: code runs only when invoked (HTTP or cloud events).
- Not stored permanently on a running server.

GOOGLE CLOUD (GCP)

General slides showing GCP services for data engineering.

AZURE

Reference to an example architecture for an insurance business: https://medium.com/...azure-services...

Databricks is a processing service, developed on top of Apache Spark, ready to work with large amounts of data. In this course we will work wth incremental data processing using Delta Live Tables (DLT) in Azure Databricks. It offers a robust and efficient way to manage and process large volumes of data by only processing the changes (deltas) since the last update

OBJECT STORAGE

AWS

- S3 – Highly durable object storage
- S3 Glacier – Archival storage

- EFS – Elastic file storage

Azure

- Blob Storage – Scalable object storage
- Data Lake Storage Gen2 – Hierarchical analytics
- Azure Files – Managed file shares

GCP

- Cloud Storage – Unified object storage
- Filestore – Managed file storage
- Persistent Disk – VM block storage

REGIONS

AWS

- 33+ regions
- 3–6 AZs per region
- 400+ edge locations
- Best in Americas and Asia

Azure

- 60+ regions (largest number globally)
- Strong in emerging markets
- 170+ CDN PoPs

GCP

- 40+ regions
- Strong global backbone
- 140+ edge locations

IDENTITY AND ACCESS MANAGEMENT (IAM)

Dimension	AWS	Azure	GCP
IAM Model	Users, groups, roles, JSON policies	Azure AD + RBAC	IAM roles + service accounts

Dimension	AWS	Azure	GCP
Policy Language	JSON	Azure Policy (JSON) + RBAC	IAM policy bindings
Federation	SAML, OIDC	Native AD integration	OIDC, SAML
Service Identity	IAM roles	Managed identities	Service accounts
Granularity	Very granular	Balanced	Fine-grained, simpler
Root Account	Root user	Global Administrator	Organization Owner
MFA	Yes	Yes (strong MFA)	Yes

PRICING AND BILLING

Dimension	AWS	Azure	GCP
Billing Granularity	Per second	Per minute–hour	Per second
Reserved Options	1–3 yrs, up to 72% off	1–3 yrs, up to 72% off	Commitments, up to 57%
Spot/Preemptible	Up to 90% off	Up to 90% off	Up to 80% off
Sustained Use	None	None	Auto discounts up to 30%
Free Tier	12 months + always free	12 months + partial free	90-day $300 credit
Cost Tools	Cost Explorer	Cost Management	Recommender
Egress Costs	High	High	High (slightly better)
Pricing Transparency	Complex	Complex	More transparent

COMPUTE SERVICES

VMs

- **AWS:** EC2 (500+ instance types)

- **Azure:** VMs (700+ sizes)
- **GCP:** Compute Engine (fewer but customizable)

Containers

- **AWS:** ECS, EKS, Fargate
- **Azure:** AKS, Container Instances, App Service
- **GCP:** GKE (best Kubernetes), Cloud Run

Serverless

- **AWS:** Lambda
- **Azure:** Functions
- **GCP:** Cloud Functions, Cloud Run

Bare Metal

- AWS: limited
- Azure: Dedicated Host
- GCP: Bare Metal Solution

NETWORKING

Virtual Networks

- AWS: VPC
- Azure: VNet
- GCP: Global VPC

Load Balancing

- AWS: ALB, NLB, CLB
- Azure: Load Balancer, Application Gateway
- GCP: Global load balancer (anycast IP)

DNS & CDN

- AWS: Route 53, CloudFront
- Azure: Azure DNS, Front Door
- GCP: Cloud DNS, Cloud CDN

STORAGE PHILOSOPHY

Category	AWS	Azure	GCP
Object Storage	S3	Blob	Cloud Storage
Storage Tiers	Standard, IA, Glacier	Hot, Cool, Archive	Standard, Nearline, Coldline
File Storage	EFS	Azure Files	Filestore
Block Storage	EBS	Managed Disks	Persistent Disks
Durability	11 nines	11 nines	11 nines

DATABASE OFFERINGS

Relational

- **AWS:** RDS, Aurora
- **Azure:** SQL Database
- **GCP:** Cloud SQL, AlloyDB

NoSQL

- **AWS:** DynamoDB, DocumentDB
- **Azure:** Cosmos DB
- **GCP:** Firestore, Bigtable

In-Memory

- **AWS:** ElastiCache
- **Azure:** Cache for Redis
- **GCP:** Memorystore

Graph

- AWS: Neptune
- Azure: Cosmos DB (Gremlin)
- GCP: No native

MANAGEMENT & OPERATIONS

Consoles

- AWS: Feature-rich, complex
- Azure: Modern, intuitive
- GCP: Clean, simple

CLIs

- AWS CLI
- Azure CLI / PowerShell
- gcloud (developer-friendly)

IaC

- AWS: CloudFormation, CDK
- Azure: ARM, Bicep
- GCP: Deployment Manager

MONITORING & OBSERVABILITY

Monitoring

- AWS: CloudWatch
- Azure: Monitor
- GCP: Cloud Monitoring

Logging

- AWS: CloudWatch Logs
- Azure: Log Analytics
- GCP: Cloud Logging

Tracing

- AWS: X-Ray
- Azure: App Insights
- GCP: Cloud Trace

SECURITY & COMPLIANCE

Compliance

- AWS: 100+ certifications
- Azure: 90+
- GCP: 70+

Threat Detection

- AWS: GuardDuty
- Azure: Defender
- GCP: Security Command Center

DDoS

- AWS: Shield
- Azure: DDoS Protection
- GCP: Cloud Armor

INTEGRATION & ECOSYSTEM

API Gateways

- AWS: API Gateway, AppSync
- Azure: API Management
- GCP: Apigee (industry-leading)

Messaging

- AWS: SNS, SQS
- Azure: Service Bus
- GCP: Pub/Sub

AI/ML & INNOVATION

AWS

- Large marketplace
- Strong partner network

Azure

- Best enterprise ecosystem
- Azure OpenAI partnership

GCP

- Best Kubernetes
- Best analytics (BigQuery)
- Strong open-source DNA

VENDOR LOCK-IN RISK

Proprietary Services

- AWS: many
- Azure: many
- GCP: fewer, more open-source alignment

Data Portability

- All: high egress costs
- GCP: slightly cheaper

MATURITY

- AWS: market leader, most mature
- Azure: strong #2, best enterprise adoption
- GCP: #3 but fastest innovations in data/AI

PERFORMANCE & SCALABILITY

- Azure supports highest networking throughput (200 Gbps).
- AWS: best overall compute variety.
- GCP: best global low-latency network.

SUMMARY & KEY DIFFERENTIATORS

Azure Strengths

- Best for enterprises
- Best hybrid cloud
- Deep Microsoft integration

- Exclusive Azure OpenAI
- Best Windows workload support

AWS Strengths

- Most mature
- Largest service portfolio
- Largest partner ecosystem
- Best for startups and general-purpose cloud

GCP Strengths

- Best for analytics and ML
- Best global network
- Best Kubernetes & multi-cloud
- Most transparent pricing

WHEN TO CHOOSE EACH

Choose Azure if:

- You're already using Microsoft ecosystem
- You need hybrid cloud
- You need enterprise-grade compliance

Choose AWS if:

- You want the broadest selection
- You want the most mature platform
- You're building SaaS or startups

Choose GCP if:

- You need heavy analytics or ML
- You want the best Kubernetes
- You want multi-cloud flexibility

COST ENGINEERING

Es diseñar y controlar una arquitectura cloud para gastar lo mínimo posible sin perder rendimiento.

En la nube pagas por todo (CPU, RAM, GB, tráfico, tiempo).
Si no controlas esto → la factura explota.

1. Cost visibility

Ver claramente quién gasta qué.
(Sin visibilidad → no puedes optimizar.)

2. Cost optimization

Reducir costes sin romper nada.
Técnicas clave:

- **Rightsizing (máquinas más pequeñas)**
- **Spot/preemptible (80% más baratas)**
- **Apagar recursos**
- **Políticas de ciclo de vida (datos fríos → storage barato)**
- **Minimizar egress (lo más caro de la nube)**

3. Cost governance

Reglas obligatorias para no gastar de más.
(Ej: etiquetas, alertas, límites, políticas.)

4. Cost modeling

Calcular antes cuánto va a costar una arquitectura usando la calculadora del cloud.

5. Cost integrity

La factura cuadra, no hay gasto fantasma, todo lo que pagas tiene sentido.

✅ **Para el examen: la frase clave**

Cost engineering es asegurar que una arquitectura cloud es eficiente, barata y sin gastos inesperados.